

Name: _____

Class: _____



JURONG JUNIOR COLLEGE

JC2 Preliminary Examinations 2018

MATHEMATICS
Higher 2

Paper 1

9758/01
28 August 2018

3 hours

Additional materials: Answer Paper
 Cover Page
 List of Formulae (MF 26)

READ THESE INSTRUCTIONS FIRST

Write your name and civics class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** the questions.
Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
You are expected to use an approved graphing calculator.
Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.
Where unsupported answers from a graphing calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.
You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

This document consists of **6** printed pages.

[Turn over

- 1** To watch the daily matches of a badminton tournament in a stadium, each spectator is to buy either a daily normal ticket for an adult spectator or a daily concession ticket for a student spectator. Tickets are purchased from either ticket booth A, B or C. The number of tickets sold and the total amount of money collected by each ticket booth are shown in the following table.

Ticket booth	Number of daily normal tickets sold	Number of daily concession tickets sold	Total amount collected
A	$5n$	n	\$5976
B	$7n$	$2n$	\$8712
C	357	51	\$5763

Find the price of each daily normal ticket and each daily concession ticket and determine the value of n . [4]

- 2** The curve C has equation $y = \frac{ax^2 + bx + c}{x + d}$, where $x \in \mathbb{R}$, $x \neq -d$ and a, b, c and d are constants.

It is given that C has stationary points at $x = 0$ and $x = -2$. The lines $x = -1$ and $y = x$ are asymptotes to C .

- (i) Write down the value of d , and determine the values of a, b and c . [6]

With the values of a, b, c and d found in (i),

- (ii) find the range of values that y can take using an algebraic method, [4]

- (iii) sketch the graph of $y = \frac{x + d}{ax^2 + bx + c}$, indicating clearly the coordinates of the points where the graph crosses the axes, the turning points and the equations of any asymptotes. [3]

- 3** (i) Find $\int \frac{x}{(4 + 3x^2)^2} dx$. [2]

- (ii) Hence find the exact value of $\int_0^{\frac{2}{\sqrt{3}}} \frac{2x^2}{(4 + 3x^2)^2} dx$. [4]

- 4 A curve C has parametric equations

$$x = \theta - \sin \theta, \quad y = 1 - \cos \theta, \quad \text{for } 0 < \theta < 2\pi.$$

- (i) Find the equation of the tangent that is parallel to the x -axis. [3]

- (ii) The normal to the curve at the point with parameter $\frac{2\pi}{3}$ meets the x - and y -axes at P and Q respectively. Show that the equation of the normal is $y = -\sqrt{3}x + \frac{2\sqrt{3}\pi}{3}$.

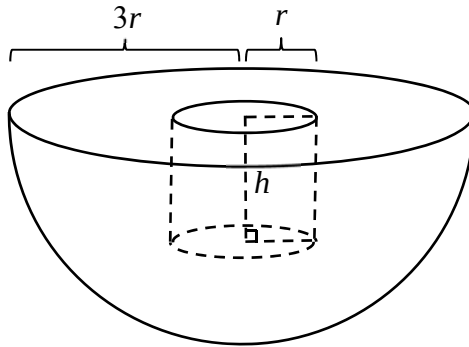
Q respectively. Show that the equation of the normal is $y = -\sqrt{3}x + \frac{2\sqrt{3}\pi}{3}$.

Hence find the exact area of the triangle OPQ . [5]

- (iii) Given that θ is increasing at a rate of 2 radians per second, find the rate of change of $\frac{dy}{dx}$ at $\theta = \frac{\pi}{3}$. [3]

- 5 [It is given that a sphere of radius r has surface area $4\pi r^2$ and volume $\frac{4}{3}\pi r^3$.]

A toy is constructed from a hemisphere with radius $3r$ cm by removing a circular cylinder of radius r cm and height h cm where $h < 3r$ as shown in the diagram below. As r and h vary, the total cost of coating the surface with a protective film on the entire toy is a constant $\$C$. The cost of coating on the flat surfaces is $\$k$ per cm^2 and that on the curved surfaces is $\$2k$ per cm^2 , where k is a positive constant.



Show that the volume, $V \text{ cm}^3$, of the toy is

$$V = \frac{117}{4}\pi r^3 - \frac{Cr}{4k}. \quad [3]$$

- (i) Find the value of r in terms of C and k which gives a stationary value of V . [2]
- (ii) Find also the ratio of the height to the radius, $\frac{h}{r}$, in this case, simplifying your answer. [2]
- (iii) Explain why it is not possible for this toy to have a stationary value of V . [1]

- 6 The position vectors of points A , B and C of a triangle are $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ respectively, relative to an origin O .

(i) By considering the area of triangle ABC , show that the shortest distance from B to AC is

$$\frac{|\mathbf{a} \times \mathbf{b} + \mathbf{b} \times \mathbf{c} + \mathbf{c} \times \mathbf{a}|}{|\mathbf{c} - \mathbf{a}|}. \quad [4]$$

R is a point on AB such that $\overline{AR} = \frac{1}{3} \overline{AB}$. S is a point on AC such that $\overline{AS} = \frac{2}{3} \overline{AC}$. $OACB$ is a kite with $OA = OB$, $CA = CB$ and OC is perpendicular to AB .

(ii) Show that $\angle SRA = 90^\circ$. [6]

- 7 (a) A function f is said to be self-inverse if $f(x) = f^{-1}(x)$ for all x in the domain of f .

The function f is defined by

$$f : x \mapsto \frac{3x+k}{x-b}, \quad x \in \mathbb{R}, \quad x \neq b, \quad \text{where } k \text{ and } b \text{ are constants.}$$

(i) Find the value of b and the set of values of k such that f is self-inverse. [3]

Using the value of b found in (i), another function g is defined such that

$$g : x \mapsto 2x-1, \quad x \neq 2.$$

(ii) Find in terms of k , an expression for $g(x)$. [2]

(b) The function h is defined as follows:

$$h(x) = \begin{cases} -4x+8, & \text{for } 1 \leq x < 2, \\ -x^2+8x-12, & \text{for } 2 \leq x < 4, \end{cases}$$

and that $h(x+3) = h(x)$ for all real values of x .

(i) Sketch the graph of $y = h(x)$ for $-4 \leq x \leq 6$, indicating the axial intercepts and endpoints clearly. [3]

(ii) Find $\int_{-\frac{3}{2}}^3 h(x) \, dx$. [3]

- 8 When a plague of locusts attacks a wheat crop, the proportion of the crop destroyed after t hours is denoted by x . In a model, it is assumed that the rate at which the crop is destroyed is proportional to $x(1 - x)$. A plague of locusts is discovered in a wheat crop when one-third of the crop has been destroyed and the rate of destruction at this instant is $\frac{1}{6}$.

- (i) Show that $\frac{dx}{dt} = kx(1 - x)$, where k is a constant to be determined. [3]
- (ii) Find the percentage of the crop destroyed two hours after the plague of locusts is first discovered. [9]

9 **Do not use a calculator in answering this question.**

- (a) Find the roots of the equation $z^2 + (i - 4)z + (6 - 2i) = 0$, giving your answers in cartesian form $a + ib$. [2]

- (b) The complex number w has modulus r and argument θ , where $0 < \theta < \frac{\pi}{2}$, and w^* denotes the conjugate of w . State the modulus and argument of p , where $p = \frac{w}{w^*}$. [2]

Given that p^6 is real and positive, find the possible values of θ . [3]

- (c) The polynomial $P(z)$ of degree 4 has real coefficients. Two of the roots of the equation $P(z) = 0$ are $z = 1 + i$ and $z = 2$.

- (i) State the number of complex roots of $P(z) = 0$, justifying your answer. [1]

- (ii) By expressing $P(z)$ as a product of linear factors, find the remaining roots of the equation $P(z) = 0$ given that $P(i) = 10 + 10i$. [5]

- 10** Building contractors are constructing a rock climbing wall at the corner wall of a gymnasium. Points (x, y, z) are defined relative to a ground anchor point at $(0, 0, 0)$, where units are metres. Support beams are laid in straight lines and the thickness of the support beams and rock climbing wall can be neglected.

The three support beams of the rock climbing wall, S_1 , S_2 and S_3 start at the ground anchor

point and go in the direction $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$, and $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ respectively. The support beams S_1 and S_2

are on the ground level. The vertices A , B and C of the rock climbing wall lie on the support beams S_1 , S_2 and S_3 respectively. The rock climbing wall lies on the plane π with vector

equation $\mathbf{r} = \begin{pmatrix} 1 \\ 5 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 3 \\ -12 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 2 \\ -7 \end{pmatrix}$, where $\lambda, \mu \in \mathbb{R}$.

- (i)** Find the cartesian equation of the plane π and hence show that the coordinates of A are $(4, 0, 0)$. [4]

One of the building safety standards stipulates that the rock climbing wall should be inclined to the horizontal ground at an acute angle not exceeding 80° .

- (ii)** Determine if this building safety standard is met. [3]

For additional stability, a fourth support beam from the ground anchor point to a point N on the rock climbing wall is laid. This support beam is the shortest in length.

- (iii)** Find the coordinates of N and the exact length of this support beam. [5]